REVISION WORKSHEET

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Practice Questions and Worked Answers

Grade 11 Advanced | Module 9 Lesson 2 | Conservation of Momentum

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: For negligible net external impulse: Σpi = Σpf. For zero net external torque: Li = Lf. Internal impulses cancel in system totals.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 38–47.

P01. Total linear momentum is conserved when the system has:

A. No internal forces B. Zero kinetic energy
C. Negligible net external impulse D. Equal masses only

Answer

C. Negligible net external impulse.

P02. In a two-object collision, the forces the objects exert on each other are:

A. Equal only if masses match B. Always in the same direction
C. Zero during contact D. Equal in magnitude and opposite in direction

Answer

D. Equal in magnitude and opposite in direction.

P03. Recoil follows mainly from conservation of:

A. Momentum B. Temperature
C. Volume D. Density

Answer

A. Momentum.

P04. When two carts stick together, their final velocity is:

A. Always zero B. The same for both carts
C. Always the original faster speed D. Independent of their masses

Answer

B. The same for both carts.

P05. In the absence of net external torque, changing an object's shape can change:

A. Its total angular momentum necessarily B. The value of Planck's constant
C. Its angular velocity while angular momentum stays constant D. Its mass necessarily

Answer

C. Its angular velocity while angular momentum stays constant.

P06. A 2 kg cart moving at +6.0 m/s collides with a stationary 4 kg cart. They stick together. Find their common velocity if external impulse is negligible.

A. 2 m/s B. 4 m/s
C. 1 m/s D. -2 m/s

Answer

A. 2 m/s. $m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$; $v_f = (2 \times 6.0 + 4 \times 0)/(2 + 4) = +2$ m/s.

P07. A stationary launcher of mass 4 kg fires a 0.2 kg projectile at +20 m/s. Find launcher recoil velocity if horizontal external impulse is negligible.

A. -2 m/s B. -1 m/s
C. -0.5 m/s D. 1 m/s

Answer

B. -1 m/s. Initial momentum = 0. Thus $Mv + mpvp = 0$; $v = -(0.2 \times 20)/4 = -1$ m/s.

P08. Two fragments initially at rest separate. Fragment 1 has mass 4 kg and velocity +6.0 m/s; fragment 2 has mass 6 kg. Find fragment 2 velocity if external impulse is negligible.

A. -8 m/s B. -2 m/s
C. -4 m/s D. 4 m/s

Answer

C. -4 m/s. $0 = m_1v_1 + m_2v_2$; $v_2 = -4(6.0)/6 = -4$ m/s.

P09. A skater changes moment of inertia from 4 to 2 kg m². Initial angular velocity is 3.0 rad/s. Find final angular velocity if net external torque is zero.

A. 12 rad/s B. 3 rad/s
C. -6 rad/s D. 6 rad/s

Answer

D. 6 rad/s. $I_i\omega_i = I_f\omega_f$; $\omega_f = (4 \times 3.0)/2 = 6$ rad/s.

P10. A 4 kg cart moving +5.0 m/s collides with a 2 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

A. 6 m/s B. 12 m/s
C. 3 m/s D. -6 m/s

Answer

A. 6 m/s. $m_1v_1 + m_2v_2 = m_1v_{1f} + m_2v_{2f}$; $v_{2f} = [4(5.0) + 2(-2.0) - 4(1.0)]/2 = +6$ m/s.

P11. Define an isolated system for a collision calculation.

Answer

A chosen system whose net external impulse during the collision is negligible, so its total momentum is conserved.

P12. Why do internal collision forces not change the total momentum of the chosen two-object system?

Answer

Their impulses are equal and opposite, so they cancel when the two momenta are added.

P13. Can momentum be conserved while kinetic energy decreases?

Answer

Yes. In an inelastic collision, momentum may be conserved while some kinetic energy becomes internal energy, deformation or sound.

P14. State conservation of angular momentum.

Answer

If net external torque is zero, total angular momentum of the system remains constant.

P15. Why must the system boundary be stated before applying momentum conservation?

Answer

It determines which forces are internal and which interactions can supply external impulse or torque.

P16. A 3 kg cart moving at +6.0 m/s collides with a stationary 6 kg cart. They stick together. Find their common velocity if external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$; $v_f = (3 \times 6.0 + 6 \times 0)/(3 + 6) = +2$ m/s.

P17. A 4 kg cart moving at +6.0 m/s collides with a stationary 8 kg cart. They stick together. Find their common velocity if external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$; $v_f = (4 \times 6.0 + 8 \times 0)/(4 + 8) = +2$ m/s.

P18. A stationary launcher of mass 6 kg fires a 0.3 kg projectile at +20 m/s. Find launcher recoil velocity if horizontal external impulse is negligible.

Answer

Initial momentum = 0. Thus $Mv + m_pv_p = 0$; $v = -(0.3 \times 20)/6 = -1$ m/s.

P19. A stationary launcher of mass 8 kg fires a 0.4 kg projectile at +20 m/s. Find launcher recoil velocity if horizontal external impulse is negligible.

Answer

Initial momentum = 0. Thus $Mv + m_pv_p = 0$; $v = -(0.4 \times 20)/8 = -1$ m/s.

P20. Two fragments initially at rest separate. Fragment 1 has mass 6 kg and velocity +6.0 m/s; fragment 2 has mass 9 kg. Find fragment 2 velocity if external impulse is negligible.

Answer

$0 = m_1v_1 + m_2v_2$; $v_2 = -6(6.0)/9 = -4$ m/s.

P21. Two fragments initially at rest separate. Fragment 1 has mass 8 kg and velocity +6.0 m/s; fragment 2 has mass 12 kg. Find fragment 2 velocity if external impulse is negligible.

Answer

$0 = m_1v_1 + m_2v_2$; $v_2 = -8(6.0)/12 = -4$ m/s.

P22. A skater changes moment of inertia from 6 to 3 kg m². Initial angular velocity is 3.0 rad/s. Find final angular velocity if net external torque is zero.

Answer

$I_i\omega_i = I_f\omega_f$; $\omega_f = (6 \times 3.0)/3 = 6$ rad/s.

P23. A skater changes moment of inertia from 8 to 4 kg m². Initial angular velocity is 3.0 rad/s. Find final angular velocity if net external torque is zero.

Answer

$I_i\omega_i = I_f\omega_f$; $\omega_f = (8 \times 3.0)/4 = 6$ rad/s.

P24. A 6 kg cart moving +5.0 m/s collides with a 3 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = m_1v_{1f} + m_2v_{2f}$; $v_{2f} = [6(5.0) + 3(-2.0) - 6(1.0)]/3 = +6$ m/s.

P25. A 8 kg cart moving +5.0 m/s collides with a 4 kg cart moving −2.0 m/s. Afterward the first moves +1.0 m/s. Find the second cart velocity; external impulse is negligible.

Answer

$m_1v_1 + m_2v_2 = m_1v_{1f} + m_2v_{2f}$; $v_{2f} = [8(5.0) + 4(-2.0) - 8(1.0)]/4 = +6$ m/s.

P26. A 2.0 kg cart at +3.0 m/s sticks to a stationary 1.0 kg cart. Find final velocity and compare initial and final kinetic energy.

Answer

Momentum gives $v_f = (2 \times 3)/3 = 2.0$ m/s. Initial KE = 9.0 J; final KE = $\frac{1}{2} \times 3 \times 2^2 = 6.0$ J. Momentum is conserved while 3.0 J becomes other energy forms.

P27. Why can a rocket accelerate in space without pushing against air?

Answer

It ejects mass backward. Momentum is exchanged between rocket and exhaust, so the rocket gains forward momentum while exhaust gains backward momentum.

P28. Two skaters push apart from rest. One has twice the mass of the other. Compare their speed magnitudes.

Answer

Their momentum magnitudes are equal and opposite. The skater with twice the mass has half the speed magnitude.

P29. A rotating platform's moment of inertia halves without external torque. Predict angular speed and discuss kinetic energy.

Answer

Angular speed doubles because $I\omega$ is constant. Rotational KE doubles; the increase comes from work done during the redistribution, so angular-momentum conservation does not imply fixed kinetic energy.

P30. A collision on a rough surface is analysed by assuming exact momentum conservation for the carts alone. What condition must be checked?

Answer

The external frictional impulse during the collision must be negligible compared with the momentum changes. If not, include it or enlarge the system appropriately.